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From Fuller´s Earth to Bleaching Earth: A Historical Note.

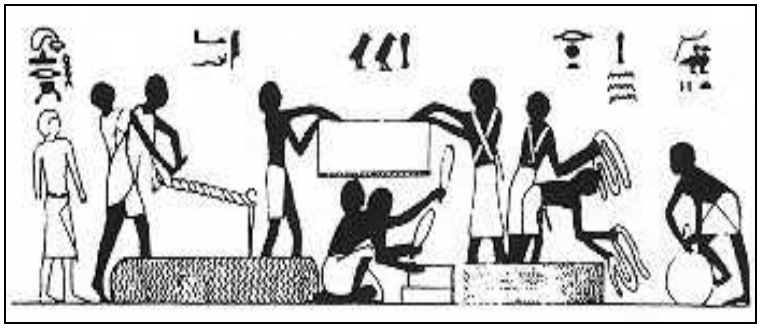
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From Fuller´s Earth to Bleaching Earth: A Historical Note.

Washing in ancient times

The washing of cloth and clothes is traced from Ancient Egypt, the Middle East, Homeric and Classical Greece, Rome and ownwards to the 20th century (ROBERTSON, 1986a; NOVELLI, 2000)



One of the most ancient historical quotation about the washing the raw clothes in fresh water is find in the Odyssey of Homer (about 800 BC):

"...(Athena talks to Nausica in her sleep) *Oh, Nausica, why did your mother give your birth so slowly? You let your clothes neglected, altough your wedding day is coming soon, when you will have to dress your limbs with airy clothes, to offer them to those who will see you under your husband´s roof...Thus, your beautiful clothes, as white as heaven, shall be rinsed into the waves...*(Afterwards, when Nausica addresses her father to ask for a chariot to take her to wash her clothes) *She told him: will you let me have the light-wheeled airy chariot ready; so that I can take those neglected and darkened clothes to the river and let it turn them white again? You deserve too much, you who own a seat among the sovereign councils, you must have neat clothes around you. Five beloved sons you can see in your house, two sons to be married and three whose youth still laughs in their unmarried faces. These ones will go to dance with fresly cleaned clothes. And those things are all in my care... Once they got to the silvery river flowing purely, and to the washing stones wet by eternal water that sheds any stains, they untied the mules... Then from the chariot, they let the clothes fall down from their arms into the river, among the waves, which all got darkened; and in large grooves they threw them, treading with*

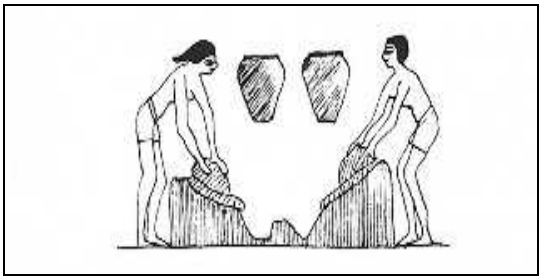
fast feet to get them cleaned. Purged and neat from any dirt, one by one they hanged them to dry on the shore, where the sea lapped the little stones..“ (NOVELLI, 2000).

The use of fuller´s earth

The English word “fuller”, as “foulon” in French or “fullone” in Italian, derives from the Latin term “fullo”, indicating a person whose job was that of degreasing and thickening clothes.

Before Homer´s description, two detergents were occasionally found out, river mud and old urine. The first must be found by washing (sheep´s) fleece in the river. The other detergent, old urine, was perhaps found by a woman who observed that the baby´s bed which got wet was whiter if washed after a few days, especially in the part which got wet with the urine.

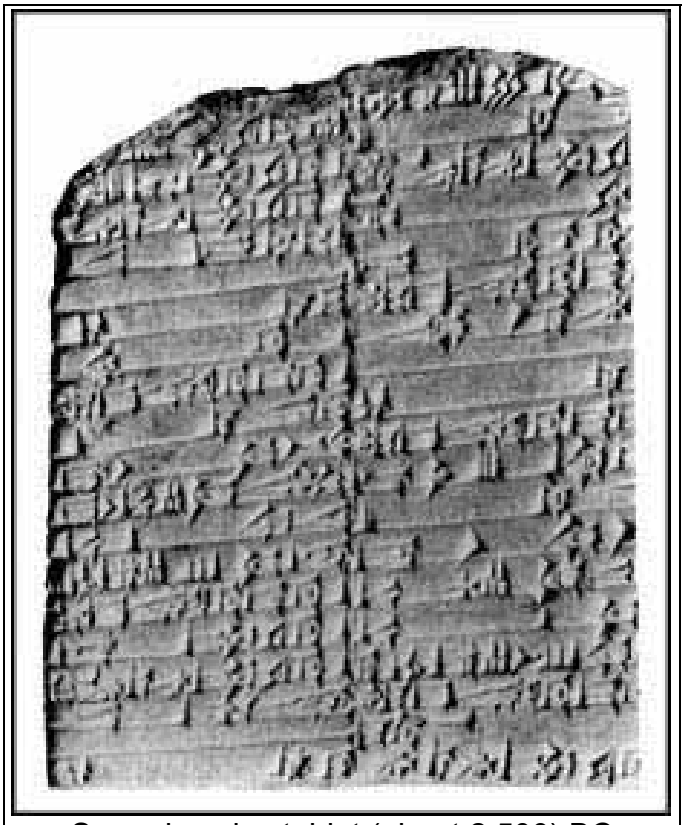
The washing of woollen cloth with fuller´s earth was certainly practised in Cyprus about 5000 BC. Sheeps were fully domesticated nearly thousand years before. There were found in houses lumps of fuller´s earth, in 1969 carbon-dated about 5000 BC. The greenish gray clay was rich in montmorillonite.



In Mesopotamia along the Tigris and the Euphrates the fuller´s earth were not of high quality. It will be possible that fleeces and cloth were fulled in the river mud in the earliest times. Later, other detergent materials were be used. Detergent materials as wood ashes or soda were used when washing in a large tub in which the fuller stirred the contents with a staff of pole because the alkali damaged the skin of the feet. Finally, a special bath (mazûru) was used to beat the cloth. In the Akkadian language fulling was called either “mazarū” or “kabāsu”. The former is named from the fulling-billet “mazûru” and its primary mining is to tramble. Respectively these words may represent fulling by hand and feet.

Ancient centres of the woollen industry were Mesopotamia and northern Syria, there was no religious ban on the wearing of woollens, in difference to Egypt which produced linen of exellent quality (ROBERTSON, 1986).

Sumerian clay tablets and a clay cylinder, found in Lagash (about 2500 B C), describe the production of cloth, fulling and washing, and also the preparation of soaps by mixing oil and potash in a volume ratio 1 : 5.5 and boiling the mixture. This soap was used for washing and cleaning cloths and clothes. The Sumerians were also aware of the preparation of medical and cosmetic scented soaps.



Sumerian clay tablet (about 2 500) BC



Hammurabi code

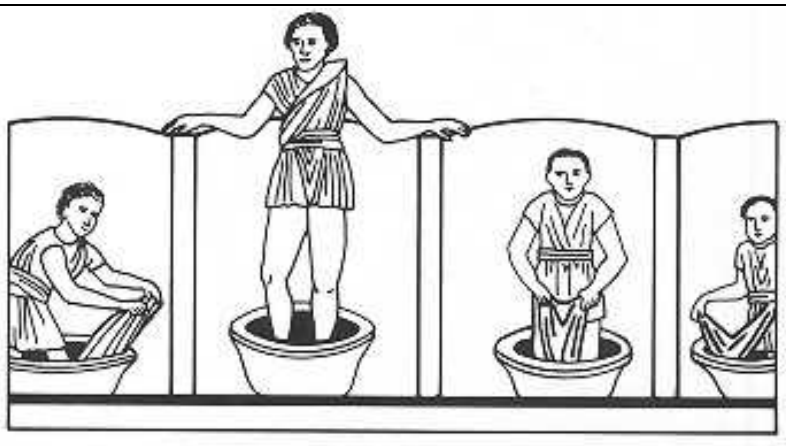
The Stockholm papyrus manuscript, found 1828 in a tomb in Thebes and dated to 3000 BC, describes a washing powder expecially for wool. This powder was composed of aqueous soda solution, Cimolian earth (clay from Kimolos, an Aegean island northeast of Melos), dispersed in vinegar, slaked lime, fine clay, and caustic potash solution.

Also vegetable compounds were added. Soap wort (Seifenkraut; Saponaria officinalis) and Asphodill (Liliaceae). Soap wort (but also Gypsophyllum (Gipskraut) and Lychnis species[1] (Lichtnelke) cause faoming and cleaning due to the presence of saponine, a vegetable glycoside which can be present in the plants up to 18 percent (LAGERCRANTZ, 1913; VOLKE, 1989; BENEKE, 1995a).

Medical prescribings and treatments with oil and water are known from Hammurabi´s time (king of Babylonia, 1728 - 1686 BC). Remedies and cosmetics were prepared from vegetable, animal and mineralic substances (fuller´s earth). Preparation of ointments and emulsions was diversified and up to 30 compounds were used. The basic compounds were animal fats (wool fat, lanoline) and vegetable oils; emulsifying agents were gumarabic, slimes, proteins, casein etc. The papyrus manuscript Ebers (dated about 1600 BC, but representing a copy of a papyrus from 2500 BC) shows more than 700 remedies and therapies and also reports the use of urine for cures (BENEKE, 1995b). In Roman times, pieces of cloth were first steeped into old urine (very rich in

[1] Lychnis is derived from the greek word meaning “lamp“ because the leaves of Lychnis coronaria were used as wicks.

ammonia) or other alkaline solutions. Then they were heavily tread upon in a large bowl or a basin containing mud, the so called “fuller´s earth”, or “creta fullonia“ in Latin. Several rinses were needed with fresh water in order to remove the fuller´s earth, which took away with it the main part of grease and dirt present into the fleece (ROBERTSON, 1986; NOVELLI, 2000).



A fuller treading the cloth, and others rinsing

Indications of the use of the fuller´s earth, old urine as detergent and the cloth washing can be detected in the Bible, particulary in the Old Testament. In the 20th century, 19 km east of Jerico a mix of earth and ashes was found and dated back to 3000 BC. In this settlement was a local washerman of the time, the name of the place has an arabian verb for “to wash”. Moreover, within the same area, a clay source was not apparentely found. Still today, Israel does not have any clay reserves of its own, as it still imports clays from Cyprus (ROBERTSON, 1986; NOVELLI, 2000).

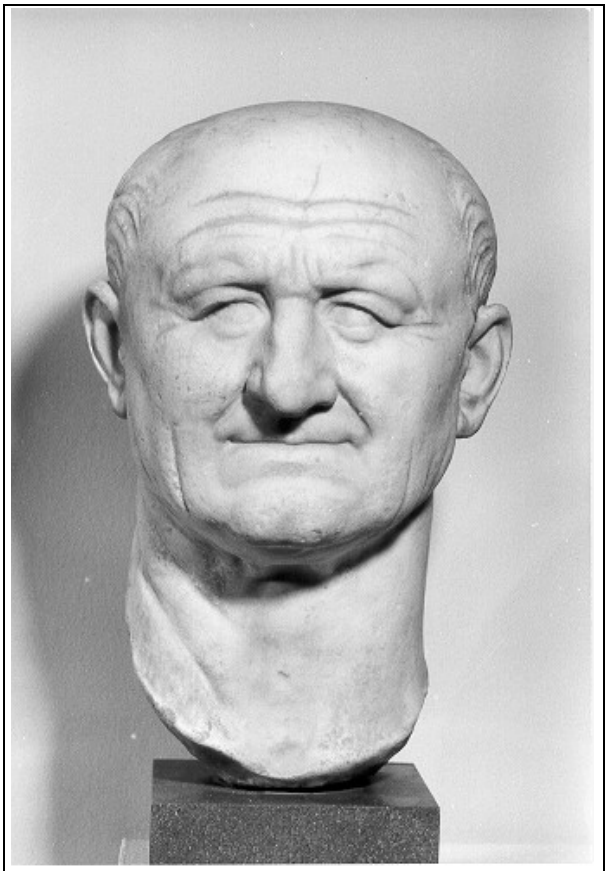
The use of urine and salpetre

As seen before urea was an important additive in washing processes. In Rome urine was collected. Large earthenware vessels “dolia curta” or “testae” were placed at street corners or in the narrow streets between the houses, and collected by the fullers when they had been filled by the passers-by. The urine of the toper had no good quality and the fullers did not like them because the nitrogen content was too low. It was just the most inveterate drinkers who had most often occasion to use the urinals. Macrobius (about 430 AD) gives a double social condemnation of the wine-bibbing dignitaries of his days:

“They sit at the gaming tables perfumed in a manner to which they have given much forethought, and surrounded by courtesans. When it is the tenth hour, they summon a slave to take them to the “comitium”, to find out what happening in the forum: what the tribunes have decreed, what they have prohibited. Then they make for the “comitium”, for fear of involment in personal responsibility for matters they have not thought about. On the way, there is not an alleyway at which they do not

stop to add to the contents of the urine vessels: their bladders beeing ever full because of quantity of wine they drink. They come with abored air into the "comitium". They order the trial to begin, the parties concerned state their case, the judge requires witness and goes out to urinate" (ROBERTSON, 1986).

Vespasian (9 - 79 AD; emporer of Rome 69 - 79 AD) collected a tax on the products of public urinals, which was used in the fullers trade at the time. When his son Titus (39 - 81 AD) complained about this method of raising money, Vespasian handed him a silver denarius. "Does this smell?" he asked. Titus answered "Pecunia non olet - money don´t stink". From this very day public urinals in French are called "Collones Vespasiennes" or short "Vespasiennes".



Vespasian, emporer of Rome

Flavius Josephus (37 - 100 AC) recounted in the "War of Jews" that asphaltene from the Dead Sea was solidified by urine. Urine was spread on the asphatene, after a certain time the liquid phase was squeezed out.

Primitive people used urine to transform the skin (animal fibers) in leather. Another application of urine was for dying. The Inuits (Eskimo) used an extract of pine tree bark in urine for red-dying of leather. In 1669, Hennig Brand (died around 1710) was looking for the philosopher´s stone but recovered the element phosphorus. During concentrating urine by evaporation in the absence of air he observed a radiation, the typical chemoluminescence of phosphorus. Guillaume François Rouelle (1703 - 1770) dedected the urea in the urine.

During the medival age urine and other nitrogen containing waste were used for formation of salpetre (nitre, KNO_3) which was required as a component in black blasting powder, a mixture of sulphur, salpetre, and charcoal.

In the 18th century the army in Bavaria used salpetre formed from liquid manure and found in the farmhouses. In the early hours representatives of the "Salpeterkommission" accompanied by soldiers, attacked the farms, removed the floor boards and scraped off the salpetre. The farmers were not much enjoyed about this procedure but found help by the catholic priests. The catholic church did not allow to demage all parts and areas in the house which were used for religious meditation. A lot of family altars were built ("Herrgottswinkel"), religious statues and pictures of saints were distributed all over the house, the "Salpeterkommission" was

often forced to restrict its activity to small areas. Is this the reason of the religiousness of the Bavarian people in the country side? (KRÄTZ, 2001).



“Hergottswinkel“ in a bavarian house

Lanolin

Wool fat (Adeps lanae anhydricus; lanolin, from latin: lana = wool, oleum = oil) is produced by the sebceous glands of sheeps and covers the skin and wool fibres to protected the sheeps against weather factors and rain. It makes the coat hydrophobic so that water trickles down. Before dying, this fat must be removed, e. g. by fulling with fuller´s earth. Raw lanolin is a greasy, foul smelling mixture of different esters. It contains more than 130 fatty acids, about 70 aliphatic alcohols (C_{14}-C_{36}) and 6 sterols. Oxidation removes the smelling and dying components so that lanolin smells more agreably. It does not easily become rancid, takes up large amounts of water and forms stable emulsions. Lanolin easily penetrates into the pores of the skin and is, therefore, used as a basic component in ointments and creams. Furtheron, lanolin is used for greasing cloths and leather and is an anti-rust agent (RÖMPP, 1990).

The Egyptian people also knew how to extract and use lanolin, because it was used in the Egypt pharmacy and cosmetics. Dioscorides (first century AD) gave in Greek a good description of how lanolin was prepared (LANGBECK, 1890):

“Wool-fat is called the grease out of filth of new shorn wool. You will prepare it in this manner: Having taken sweet, newly shorn wool, not treated with soap wort (Saponaria officinalis), wash it in hot water, at the same time squeezing out all the filth; having put this in wide-mouthed vessel, pour water over it, and agitating it violentely until froth is formed, pour it out again of the vessel; or agtiate it vigorously with a stick until much scum and filth collects; therupon sprinkle sea water over it,

and when the fat swimming on the top has set, put it into another earthenware vessel; pour more water into the tub, stir it up again, sprinkle the froth with sea water and take it out again, and do so until after the fat has been exhausted, [when] no more froth gathers. Smooth the collected wool fat with the hand, and if there is any filth remove it forthwith, squeezing out the first water and pouring on fresh and agitating it with the hand until, applied to the tongue, it does not bite but has an astringent taste and looks brilliant and pure and white. In this condition store it away in an earthenware vessel. But all must be made in hot sunshine. Some treat the fat by washing it in cold water, rubbing it with hands as woman [do with] wax, thus rendering it whiter. Others, after washing the wool and squeezing out the grease, boil it together with water in a cauldron over a slow fire, and having gathered the fat which swims on the top they wash it with water, as has been said, and straining it into an earthenware dish containing hot water, and concerning it with a linen cloth they place it into the sun until it becomes sufficiently thick and white. Some pour off the first water after a lapse of two days, and pour on fresh. Better is that which has not been treated with soap wort and is smooth, smelling of the oily wool, washed with cold water in a shallow dish, and having been bleached and containing in itself no hard or solid [particles], as for instance that which is adulterated with wax or tallow. It possesses a power to close up wounds; to soften, especially with melilot and butter" (ROBERTSON, 1986).

Langbeck wrote that Nicolas Culpeper had published in 1653 "The manner of preparing Oesypus", which was based on Dioscorides (first century AD). In 1882 Braun and Liebrich took out a patent for the purificication of wool fat, which, mixed with water, they called "lanolin". The product was dirty yellow and still contained free fatty acids from the soap. Langbeck and Ritsert took out a German patent in 1886 to purify wool fat. At this time, LANGBECK (1890) published the story of purifying wool fat from Dioscorides (HUSEMANN, 2002).

Dying

Cleaning and decreasing of wool and cloths was an important condition for dying. In ancient Egypt the job of washer had a special importance, these people were not found in lower social classes. A few hieroglyphic texts report washing operations.

More than 4000 years ago the Egyptians not only used vegetable dyes, animal dyes (purple) and a few mineral pigments for dying wool and cloths but also knew the complicated procedure of dye vatting with indigo.

Wool was often first dyed blue with indigo and then dyed again with some other dye like madder (Krapp); the result yielded a purplish colour. A recipe for this re-dying with madder can be found int the earliest alchemical papyrus. Ash is scattered over the indigo-dyed wool, and then trodden with the feet. Then the wool was rinsed in water and clay (fuller´s earth) and pickled in alaun. The madder was dissolved, together with some flour of beans to make the water softer. The treated

wool was dyed for the second time. In the last century fuller´s earth has been used for preventing indigo-dyed uniforms from bleeding (ROBERTSON, 1986).

Saffron-dyed bandages were found on Egyptian mummies of the 12th dynasty (about 2500 BC) (BENEKE, 1995c).

The ancient method of preparing woad was by fermentation in urine. Dyers did not always escape the odium experienced by fullers. In an Egyptian papyrus (about 2000 BC) it was said of a dyer: *"His hands stink, they have the odour of rotten fish, and he abhorreth the sight of all cloth"*.

Kimolian and other earths

In Greek theatre Aristophanes (c. 445-385 BC) was the first to mention Cimolian earth. In the "Frogs" (c. 405 BC) is written:

"That Cleigenes shifty and small,
the wickedest bathmann of all
Who are lords of the earth, which is brought
from the isle of Kimolos, and wrought
with nitre and lye into soap
not long shall be next us, I hope".

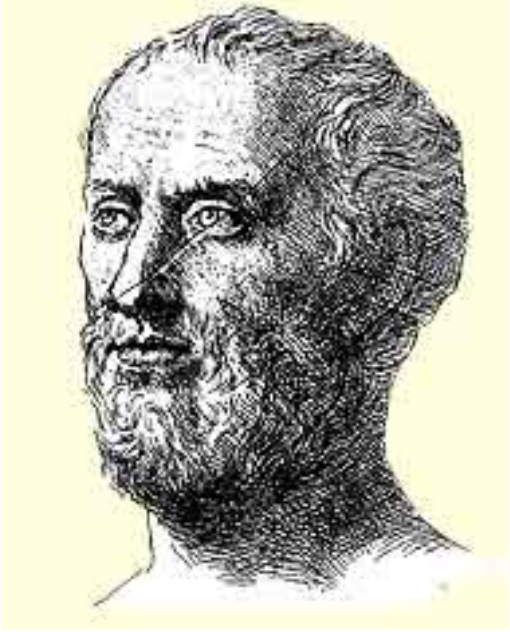
Dioscorides confirms the value of this mixture that the best soap is made with nitre and kimolian earth. The product was a body detergent and not a fulling agent. The mixture, to be more precise, consisted of sodium montmorillonite, an impure form of sodium carbonate, and plant ash. The products of the reaction being sodium and potassium montmorillonite and calcium carbonate. Kurd Endell (1887 - 1945) and Ulrich Hofmann´s (1903 - 1986) process for transforming calcium montmorillonite into sodium montmorillonite was a German patent by S. Erbslöh, 1934 (British patent 447 710; 1935):

The calcium bentonite was dispersed with water and 2-2.5 % (w/w) sodium carbonate. The diluted slurry was allowed to stand for a certain time before the product was filtered, washed and dried. The dried material consisting of sodium montmorillonite and calcium carbonate become thixotropic when 15-20 % (w/w) water were added. As drying was expensive, Ulrich Hofmann and Kurd Endell changed the process 1936. The sodium carbonate was added to the calcium bentonite containing 20-30 % moisture. By intense mixing (grinding) a product very similar to the product of the wet process was obtained. Hofmann and Endell took out a German and British patent (458 240) in addition 447 240. Aristophanes was cited to legal counsel (ROBERTSON, 1986a; NOVELLI, 2000).

Plant dyestaff in ancient times

(VOLKE, 1988; BENEKE, 1995c)

Color	Name English (German)	Latin Term	Family	Organ	Dyestaff
Red	Madder (Färberröte, -wurzel, Krapp)	Rubia tinctoria	Rubiaceae	Roots	Alizarin, Dihydroxy-anthraquinone, Purpurin
Red	Safflower False Saffron (Saflor, Falscher Safran, Färberdistel)	Carthamus tinctorius	Compositae	Blossom	Safflor red, Carthamin
Blood red	Alkanet (Alkannastaude)	Alkanna tincoria	Boraginaceae	Roots	Alkannin,
Dark red	Garnet, Inkberry, Pigeonberry, Pokeweed, Scoke (Kermesbeere)	Phytolacca americana	Phytolaccaceae	Sape of berry	Phytolacca red (Kermes red)
Orange red	Henna, Mignonette-Tree (Henna)	Lawsonia inermis	Lythraceae	Leaves	2-Hydroxy-1,4-naphtoquinone
Gold yellow	Greater celandine, Swallow-wort, (Schöllkraut, Schöllwurz)	Chelidonium Majus	Papaveraceae	Roots	Chelidonine
Yellow	Saffron (Safran)	Crocus sativus	Iridaceae	Stigma	Crocin, Polychroit
Yellow	Wild Mignonette (Färberkraut, Wau)	Reseda luteola	Resedaceae	Green plant sections	Luteolin
Yellow	Dyer´s Greenweed (Färberginster, Gilbkraut)	Genista tinctoria	Papilionaceae	Blossom	Glucosido-genistein
Blue	Litmus lichen (Lackmus-flechte)	Rocella tinctoria and further species	Rocellaceae	Total lichen (fermentation)	Phenoxazon-derivative
Dark blue	Dyer´s Woad (Färber-Waid)	Isatis tinctoria	Cruciferae	Leaves (fermentation)	Indigo, Indigoblue
Dard blue	Indigo (Indigostrauch)	Indigofera tinctoria Indigofera suffruticosa (= Indigofera anil)	Papilionaceae (Leguminosae)	Branches	Indigotin, Indigoblue
Purple	Orchella Weed (Orseilleflechte)	Rocella tinctoria	Rocellaceae	Total lichen (fermentation)	Phenolorcinol, Orcein



Theophrastus

Theophrastus (372 - 287 BC), born at Eresos on the island of Lesbos, was the son of Melantas, a fuller. He wrote one of the earliest known scientific works “On Stones” (about 314 BC) dealing with stones, industrial minerals and artificial products derived therefrom. Theophrastus wrote in Greek:

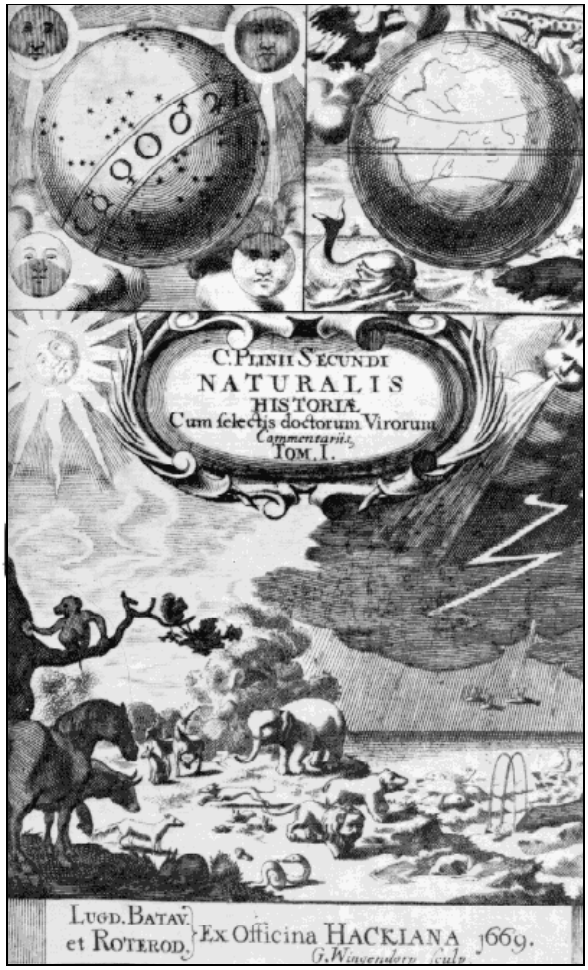
„...there are, roughly speaking, three or four natural earths which are useful as well as unusual, namely the Melian, the Cimolian, the Samian, and finally besides these the Tymphaic, otherwise known as gypsum...”

Theophrastus says nothing more about the Kimolian earth. It must be well known. Melian earth, used in painting pictures, was rough, not greasy and of loose texture, and probably the naturally-occurring fine silica, known now as “Milowite”. Dioscorides adds that it makes a sound when rubbed between the fingers. Samian earth, probably a kaolin, was extracted from a narrow sloping vein. The lower quality material was nearest the headwall and footwall; and the best quality, selectively mined, bore the trade name “The Star”, either because it was stamped with this figure or because it was highly reflecting. According to Dioscorides it clings closely to the tongue; this and its use in plasters and salves also suggest that the clay was kaolin. Deposits of kaolin, no doubt from a more accessible site, were worked in ancient times in the island for making ceramic ware. The “Star” grade of earth, according to Theophrastus, was used mainly, if not entirely, for treating cloaks, presumably because its white crystallites gave the cloth a clean and white appearance as in pipe-clayed webbing or china clay-coated art paper (ROBERTSON, 1958, 1986).

Pliny the Elder (23/24 - 79 AD) explicitly quotes the so-called “Terra Cimolia” in his “Naturalis historia” (77 AD). For this work he has used over 2000 works from elder authors. He refers to a quotation of a Greek physician from Anazarba, Dioscuri Pedanio, who lived in the first century AD near Tasso, Cilicia. Pliny describes deposits of Kimolos clay, distinguishing them in two types, a white one and a reddish one. He cites other varieties in Thessaly, Lycia and near Bubo. He wrote also about a Sardinian quality, used exclusively for white garments, not for coloured ones. It must be known that until now an exellent bentonite quality is found on the island of Kimolos.

Pliny the Elder wrote in his “Naturalis historia”: *“Cimolean creta has another use, too - the whitening of clothes. The kind which comes from Sardinia, and is called Sarda, being employed only for white garments and useless for coloured ones, is the least valuable of all kinds in Cimolia. The Umbrian sort and the one they call Saxum*

are more highly prized. Saxum has the property of increasing in bulk when soaked in water and is purchased by weight, whereas Umbrian is sold measure. The only use of Umbrian creta is for imparting lustre to garments.



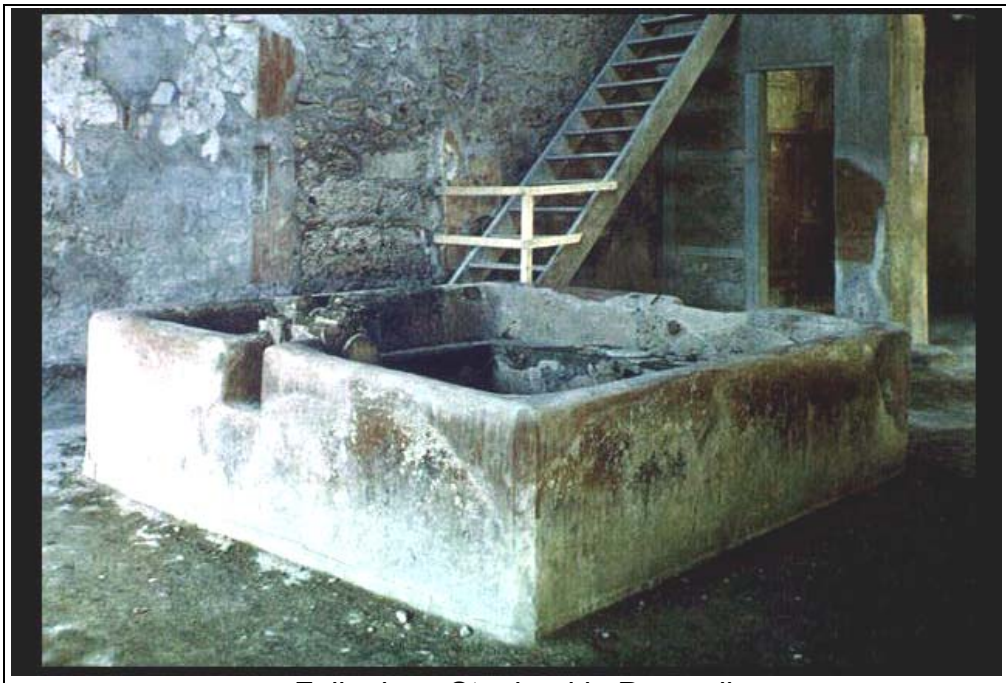


Gaius Plinius Secundus

This routine, therefore, must be observed: the garment is first washed with Sarda, them fumigated with sulphur, and finally scoured with Cimolian creta, provided that the dye is fast, for bad dye is detected by this procedure and blackens, and its colour is destroyed by sulphur. On the other hand, the valuable fast colours are softened by Cimolian creta, with freshens and brightens hues which have been dulled by sulphur. If thegarments are white, it is better to follow the sulphur by saxum, but this cannot be used with colours. Instead of Cimolian earth, the Greeks use the gypsum of Tymphaea". The fumigation with sulphur was to bleach the cloth and was a standard method early last century in Italy (ROBERTSON, 1986, NOVELLI, 2000).

Gaius Plinius Secundus, Pliny the Elder, died on sulphur dioxide poisoning when Pompeii was buried in the ash of Mount Vesuvius in 79 AD. He went to investigate the type of violent volcanic eruption which bear his name (Plinian eruptions), and to see if he could help in any way.

Excavations at Pompeii have revealed better than any literary accounts the art of the fuller. Pompaii was at the time of its overwhelming one of many cloth manufacturing towns in Italy, and prominent as a center of production of finer quality materials. In 1929 four large "fullonicae" were excavated in Pompeii. One belongs to the freedman Stephanus, whose installations are the best preserved.



Fullonicae Stephani in Pompaii



Paper mill in Nuremberg (1391)

Fulling millers in the medival age

The mills (for milling grain) and paper mills (to chop up rags, the basis material of paper before the introduction of cellulose) were invented in the medival age. The first paper mill was built in Nuremberg in 1391. The mills were water-driven. A few years later, in 1408, a windmill was constructed in the area which is now Netherland. This windmill was used to pump water from inland areas into the sea. Mills were also built for washing and combing wool and cloths. The profession of “fulling miller” came into being, a profession which disappeared in the time of industrialisation. This miller produced loden wollen fabric in the following way: After shearing the sheeps, cleaning and removing the fat, the wool was combed and weaved in wooden vats. The slack fabric was soaked in a dispersion of soap and fuller´s earth. The textile was fulled with water-driven wooden mallets to felt and condense the fabric. Finally the cloths were lifted and lowered by a water-driven shaft.

The English fuller´s earths



Fuller´s Earth Cream

In Great Britain the fuller´s earth was not only used for fulling cloths but also in pharmaceutical and cosmetic applications in form of powders and ointments, in particular in the 17th century after detecting deposits of fuller´s earth. These deposits were located in a circular around London, in Surrey around Reigate, in Kent at Madstone, and in Bedfordshire near Woburn. The English fuller´s earth was particularly effective. At the continent, deposits existed in 1929 in Bohemia, Rhône area (France), Italy (western side of the Apennine Mountains) and Transsilvania (Romania). Large deposits were found in the Caucasus near Kutais in Georgia.

The process of fulling was described 1847 in a Handbook of Technology:

"The important component for fulling is rotted urine containing ammonium carbonate. Further impartent additives are soap and fuller´s earth, a special sand-free clay. Clay and urine are mixed, and the soap is dissolved in the mixture. Other fullers start with fulling in urine and in a subsequent step with aqueous soap solution" (KNAPP, 1847).

Because of the need of urine, textile factories were often placed near barracks. Today fulling is no longer in use; the wool is now cleaned with special washing powders.

Bleaching earths

An American traveller in the Middle East observed that people used clay to lighten olive oil. Fairnanks & Co. in Chicago tested the clays to bleach cotton seed oil. The best results was obtained with the English fuller´s earth. This earth was then used for bleaching vegetable oils in USA since 1878, and became the most important fuller´s earth on the European and American market. However, about ten years later, John Olson found clays in Arkansas (in 1891) which were similar to the English fuller´s earths. When, a few years later, large deposits were found near Quincy in Florida, the monopoly of the English fuller´s earths was broken.

The Florida earth (about 33 000 tons in 1912) was mainly used for oil refinery. Vegetable and animal oils were purified with English fuller´s earth (about 20 000 tons in 1912) up to the First Word War.

Production of Fuller´s Earth by decades (1854-1970) in United Kingdom
(ROBERTSON, 1986)

Period	Thousand tons	Average tons Per annum
(1854-1860)	85	12140
1861-1870	81	8100
1871-1880	Not known	not known
1881-1890	154	15400
1891-1900	178	17800
1901-1010	242	24200
1911-1920	303	30300
1921-1930	239	23900
1931-1940	203	20300
1941-1950	446	44600
1951-1960	790	79000
1961-1970	1347	134700
Total:		
1854-1970	4068	

Before bleaching with fuller´s earths, all types of oil, fats, and waxes were bleached with very different materials, in particular residues from melts for production of prussiate of potash and animal charcoal. These expensive materials adsorbed the dying components in oil but also bound large amounts of the oil it self (up to 2.5 g oil per g adsorbent !).

Production and use of bleaching earths increased in USA, and 1924 allmost all material was mined and produced in the South-States of USA. The fuller´s earths were dried, milled, and divided into several fractions, but unfractionated material was also used.

As only a few deposits of fuller´s earths were found in Germany (Saxonia, Westerwald, Silesia), several thousand tons of American and English fuller´s earths were imported.

Since 1924 modified clays from Death Valley, bentonite from California, Nevada (Utah), Wyoming (Montana) etc. began to threaten the market of the fuller´s earths. Following an American patent these bentonites were treated with sulfuric acid and became superior to natural fuller´s earths (ECKART AND WIRZMÜLLER, 1929).

Bleaching earth industry in Bavaria

The bleaching earth industry in Bavaria goes back to 1904. During ground excavation one of the future founders of the Pfirschinger Mineralwerke in

Kitzingen/Main discovered a bentonite deposit. The white and lightly bentonites were used as white pigments (Pfirschinger Weißerde). The more yellow types were marked in 1906 as bleaching earths.

The Pfirschinger Mineralwerke Gebr. Wildhagen und Falk in Kitzingen/Main developed in 1906/07 the acid treatment of the bentonites to increase the bleaching ability, and a production facility was built in 1908. An amount of 1000 tons of this bleaching earth was selled in 1909 under the trade name “Frankonit“.

An other bentonite deposit was discoverd by von Delling in Kronwinkl near Moosburg during ground exacavation for a villa. This white material could not be used for ceramics, neither for production of cement and in the chemical industry. It was only used as binder (“Fixiererde“) in paints. The Erdwerke Kronwinkl Franz Schmid & Co. were founded in 1906. Shortly later the name was changed in Tonwerke Moosburg A. und M. Osterrieder, today Süd-Chemie AG.

Dr. E. Theobald (partner of this company) invented in 1909 the acid-activation of bentonite by hydrochloric acid without knowing the procedure of the Pfirschinger Mineralwerke. Since 1909 this bleaching earth was and is marked as “Tonsil“ (ECKART AND WIRZMÜLLER, 1929).

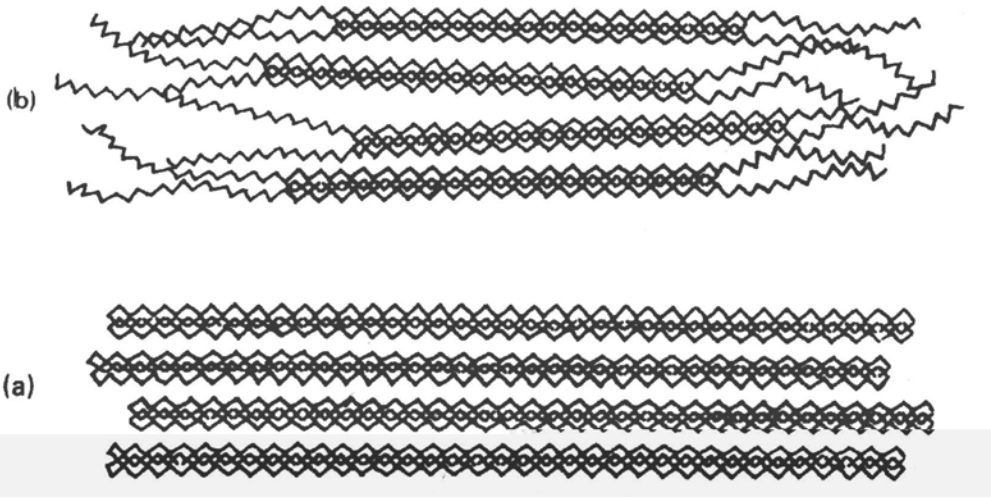
The production of bleaching earth from bentonites comprizes the reaction with acids at boiling temperature, washing with water, drying and milling. During the acid attack the interlayer cations are replaced by protons and the octahedral cations (Al^{3+}, Mg^{2+}, $Fe^{2,+}$, Fe^{3+}) are dissolved. The products are X-ray amorphous, of high SiO_2 content, and show a high specific surface area and micro- and mesopores (FAHN, 1973). The bleaching power depends on the activation procedure (acid concentration, temperature, pressure, time) (FAHN, 1973).

As a patent for bleaching earth preparation was not submitted, the procedure was kept secret, neverless, it became generally known within a short time. The bleaching earth preparation was promoted by the First World War. Because of insufficient washing, the earths were often of low quality. The materials retained some amounts of acid which reduced the bleaching power.

The Pfirschinger Mineralwerke in Kitzingen/Main tried in 1919 to receive supplementary protection for their procedure by a patent which was granted by D. R. P. 339 919. Compared with a few patents before (Ludwig Kern D. R. P. 304 076; Josef Kohlndorfer D. R. P. 305 452; Westfählische Mineralölwerke Schmitz und von der Heide D. R. P. 315 467), D. R. P. 339 919 was of much more practical importance. In the patent the procedure is described in the following way:

“It is known that filter and clearing materials such as bone char, resides of the preparation of prussian of potash and clarifying earths are washed with diluted inorganic and organic acids before using. Washing removes impurities and by dissolving certain compounds increases the effectiveness. Experiments of the inventor have shown that the clarifying and bleaching power of bleaching earths for oils, fats, waxes, paraffins, mineral oils etc. mainly depends on the amounts of hydrated silica in the bleaching earth. Reacting natural silicates with sufficiently

strong acids leads to formation of this silica which strongly enhances the clarifying and bleaching effect. Further experiments have shown that hydrochloric acid can be fully or partially replaced by other acids. Silicates have to be reacted with the acid until they are decomposed to silica to a large extend. There is also the possibility to dissolve the silicates with aqueous or melted alkaline compounds (for instance soda) and in a following step to precipitate the silica with acids. The temperature at which the silicates are decomposed depends on the agents used and the intented degree of decomposition. It also depends on the reacting component and the degree of decomposition if the reaction is carried out in air, at pressure or in vacuum. The obtained bleaching earth may be washed with water and dried. A special procedure is the use of alumosilicates like clay, kaolin etc. Their content of natural silica can strongly enhanced by the described method. One finds in the journal "Der Seifenfabrikant" (1908, page 942) the „bleaching earths are treated with concentrated acids like concentrated hydrochloric and sulfuric acid". However, the main process of this procedure is the removal admixtures. Even if a certain decomposition of the material is observed, the experiments reported in the journal reveal that the bleaching effect is not influenced in a way as described in the present procedure. As reported in the journal the bleaching power of the earth treated in concentrated acids can even be reduced in comparison with the untreated earth."



(a) silicate layers of montmorillonite
(b) disorientation of acid-treated montmorillonite layers with amorphous silica in edge regions of crystallites (FAHN, FENDERL, 1983)

The patent claim of D. R. P. 339 919 is: *„Verfahren zur Erhöhung der Entfärbekraft von Silikaten für Fette, Öle, Wachse, Paraffin, Mineralöle und dergl., dadurch gekennzeichnet, daß die Kieselsäure natürlicher Silikate durch aufschließende Behandlung, insbesondere mit genügend starker Salzsäure, ganz oder größtenteils in Kieselsäurehydrat übergeführt wird."*

(Procedure for increasing the decolourising power of silicates for fat, oils, waxes, paraffin, mineral oils and others, where the silicic acid ("Kieselsäure") of natural silicates fully or partially is transformed by decomposition, especially with sufficiently concentrated hydrochloric acid, into hydrated silica ("Kieselsäurehydrat")).

World productions (in thousands tons) of bentonite in the decade 1984-1993

(NOVELLI, 2000)

	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993[1]
Main Regions										
EU	1159	1289	1921	1892	1358	2063	1559	1728	1697	1708
Other Europe	-	-	148	154	125	129	103	85	56	51
Africa	143	83	161	204	213	215	209	206	170	166
Asia	552	621	663	708	678	727	804	847	822	814
America	3733	3554	3101	3042	3408	3563	3977	3911	3331	3468
Oceania	46	35	43	30	37	36	36	35	35	35
Total MEC	5633	5582	6037	6030	5819	6733	6688	6812	6111	6242
Other NMEC	3194	3205	3241	3253	3226	3229	2957	2604	2181	1785
World Total	8827	8787	9278	9283	9045	9962	9645	9416	8292	8027
Main Countries										
Argentinia	126	147	146	109	174	119	123	136	82	90
Brazil[2]	201	236	229	217	147	118	180	130[1]	130[1]	130[1]
Germany[3]	-	-	179	167	197	515	577	583	581	575
Greece	314	506	346	404	444	601	554	559	552	544
Italy	304	297	300	313	319	234	228	385	360	300
Japan	410	462	478	469	455	526	549	554	534	519
Mexico	268	268	137	130	164	124	145	145	136	127
Romania[1]	180	180	185	180	180	175	150	150	120	120
Spain	73	90	115	103	104	143	151	150[1]	150[1]	150[1]
USA	3119	2898	2552	2546	2871	3112	3474	3432	2954	3093
USSR/CIS[1]	2880	2890	2900	2900	2900	2900	2700	2400	2000	1600

Units: [1]: estimate values; [2]: beneficiated; [3]: includes the former East Germany

D. R. P 339 919 provided the Pfirschinger Mineralwerke the monopoly for preparation of an effective bleaching earth. Nevertheless, more and more companies developed which tried to keep secret their procedure but it was evident that their procedures were similar to the patented procedure or even identical. A long patent

controversy developed, and at June 23, 1924 the “Reichspatentgericht“ decided that D. R. P. 339 919 has to be considered not granted (ECKART AND WIRZMÜLLER, 1929).

Bleaching earths are world-wide produced and used for refining, decolourisation, clarifying and stabilisation of vegetable and animal oils and fats, of mineral oils, fining of wine, must, juices, stabilisation of beer, clarifying of saccharose syrup, catalysator for oil cracking, filters, adsorbents of radioactive compounds, binding of oil on water.

Organic activated bentonites

John W. Jordan Jr. (1912 - 2001) was the first fellow who developed a research group on organophilic bentonites (1949), today known as bentones (JORDAN, 1949; JORDAN, HOOK, FINLAYSON, 1950; JORDAN AND WILLIAMS, 1954). Besides alkali activated and acid activated (bleaching earths) organophilic bentonites are used in many applications.

Application of acid-activated (a), soda-activated (s), organic-activated (o) and raw (r) Bentonites

Areas of applications	Uses	Type of activation
Agriculture, Horticultere	Soil improvement; Composting	r, s r, s
Building industry	Supporting dispersions for cut-off diaphragma wall Constructions, Shield tunneling Subsoil sealing; Antifricton agents for pipejacking and shaft sinking	r, s r, s r, s r, s
Ceramics	Plasticizing of organic masses, improvement of strength; Fluxing agents	r, a r, a
Chemical industry	Sulphur production: refining, decolourising, bitumen extraction; Catalysts Catalyst supports; Supports for insecticides, fungicides; Fillers; Dehydrating agents; Adsorbents for radioactive materials	r, a r, a, s r, a, s r, a, s, o r, a, s, o r, a, s r, a, s
Cleaning industry	Regeneration of organic fluids for dry cleaning; Polish and dressings; Additives for washing and cleaning agents and soap production	a r, s r, s
Environment	Forest and water conservation: Fire extinguishing powders; Binding agents for oil on water; Animal husbandry: manure treatment, cat litter; Water and waste water purification; Sewage sludge pellettising; Barriers	r, a r, a r, s r, s r, s r, a, s r, a r, s

Foodstaff industry	Refining, decolourising, purification and stabilisation of vegetable and animal oils and fats; Fining of wine, must, and juices; Beer stabilisation; Purification of saccharine juice and syrup	r, a r, a, s r, a, s r, a, s
Foundries	Binding agents for special moulding sands; Binding agents for synthetic moulding sands, core sands; Binding agents for anhydrous casting sands, for thickening blackwashes	r s o
Mineral oil industry	Refining, decolourising, purification and stabilisation of mineral oils, fats, waxes, and paraffines; Drilling fluids; Thickening of greases; Catalysts for oil cracking	r, a r, a, o o r, a
Paints and varnish industry	Thickening, thixotroping, stabilising, and antisettling agents for paints and varnishes; Coating materials; Sealing cements, waxes, adhesives	s, o s, o s, o
Paper making	Pigment and colour developer for carbonless copying paper; Adsorption of impurities in circulation water; De-inking in waste-paper recycling	r, a, s r, a, s r, a, s
Pharmaceutical industry, cosmetics	Bases of creams, oitments and cosmetics; Drug carriers, powders, ointments, tablets; Stabilisator of emulsions; Odour control, liquid adsorption	r, a r, a, s r, o r, a, s, o
Tar exploitation	Emulsification and thixotroping of tar-water emulsions; Tar and asphalt coatings	a, o a, o

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